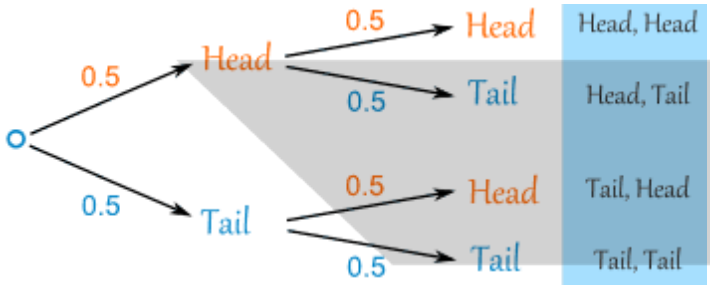
1. Probability

1.1. Basic Rules

- All probabilities lie between 0 and 1
- $P(A) =$ The probability of event A

$$\mathrm{P(A)} = \frac{\text{Number of equally likely outcomes}}{\text{Total number of equally likely outcomes}}$$

- $P\left(A'\right) = 1 - P(A) =$ The probability of not A
- To simplify a question, represent info in a tree diagram:



{S08-P06} Question 7:

A die is biased so that the probability of throwing a 5 is 0.75 and the probabilities of throwing a 1, 2, 3, 4 or 6 are all equal. The die is thrown thrice. Find the probability that the result is 1 followed by 5 followed by any even number.

Solution:

The probability of getting a 1

$$1 - 0.75 = 0.25$$

5 numbers $\therefore 0.25 \div 5 = 0.05$

The probability of getting a 5 $= 0.75$

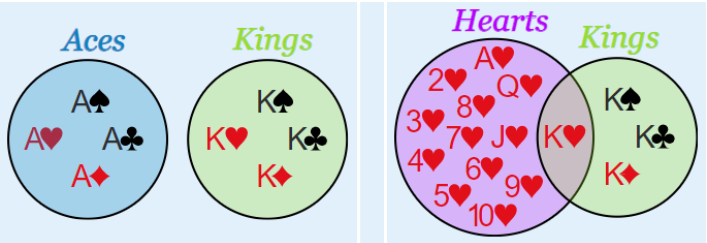
The probability of getting an even number can be 2, 4 or 6 $\therefore$

$$0.05 \times 3 = 0.15$$

Total Probability

$$0.05 \times 0.75 \times 0.15 = 0.00563$$

1.2. Mutually Exclusive Events



Aces and Kings are **Mutually Exclusive** (can't be both)

Hearts and Kings are **not** Mutually Exclusive (can be both)

- 2 events which have no common outcomes or can't happen at the same time.
- Examples of MEEs:
  - Looking Left & Looking Right.
  - Coin toss: Heads & Tails.
  - Cards: Kings & Aces.
- Examples of *not* MEEs:
  - Cards: Kings and Hearts. (we have Kings of Hearts)
  - Students: People who study French and Spanish (some can study both)
- Rules of MEEs:

$$P(\text{A and B}) = 0$$

In other words, the probability of both events happening is 0 because they can't occur at the same time.

$$P(A \text{ or } B) = P(A) + P(B)$$

In other words, the probability of A or B happening is just their individual probabilities being added together. *Example:*

P(Get King AND Queen) = 0

P(Get King OR Queen) = 1/13 + 1/13

- Rules of *Not* MEEs:

$$P(\text{A and B}) \neq 0$$

Because events can happen at the same time, the probability of both events happening is not 0.

$$P(\text{A or B}) = P(A) + P(B) - P(\text{A and B})$$

When adding P(A) & P(B), we counted the "middle part" twice, so we must subtract it.

Example:

P(Get Kings or Hearts) = 4/52 + 13/52 – 1/52

1/52 comes from the king of hearts

1.3. Conditional Probability

- Calculation of probability of one event given that another, connected event, had occurred.
- Conditional Probability formula:

$$P(B|A) = \frac{P(\text{A and B})}{P(A)}$$

Or, in another form:

$$P(A \text{ and } B) = P(A)\, P(B|A)$$

"Probability of event A and event B equals the probability of event A times the probability of event B given event A has occurred"

Example:

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$$P\left(Get\ 2\ kings\right)=\frac{4}{52}\times\frac{3}{51}$$

Both events are "Get a king" but since one king has been picked up, the next probability has changed. In other words, $\frac{3}{51}$ *is* $P(Get\ a\ king \mid Get\ a\ king)$ *, the probability that you get a king given that you've already gotten one.*

{S07-P06} Question 2:

Jamie is equally likely to attend or not to attend a training session before a football match. If he attends, he is certain to be chosen for the team which plays in the match. If he does not attend, there is a probability of 0.6 that he is chosen for the team.

1. Find the probability that Jamie is chosen for team.
2. Find the probability that Jamie attended the training session, given that he was chosen for the team.

Solution:

Part (i)

Probability attends training and chosen

$$0.5\times 1=0.5$$

Probability does not attend and is chosen

$$0.5\times 0.6=0.3$$

Total probability

$$P\left(\text{Chosen}\right)=0.3+0.5=0.8$$

Since we are looking for $P\left(\text{Chosen}\right)$ *, we have to find every possible way (every possible probability) that Jamie is chosen, then sum them up.*

Part (ii)

This is a conditional probability question, and we have to look for $P\left(\text{Attends} \mid \text{Chosen}\right)$ *.*

$$P\left(\text{Attends}|\text{Chosen}\right)=\frac{P\left(\text{Attends and Chosen}\right)}{P\left(\text{Chosen}\right)}$$

$$P\left(\text{Attends}|\text{Chosen}\right)=\frac{0.5}{0.8}=0.625$$

Old Question Question 7:

Events A and B are such that $P\left(A\right)=0.3,\ P\left(B\right)=0.8$ and $P\left(\text{A and B}\right)=0.4$. State, giving a reason in each case, whether A and B are

1. Independent
2. mutually exclusive

Solution:

Part (i)

A and B are not mutually exclusive because:

$P(A\ and\ B)$ does not equal 0

Part (ii)

A and B are not independent because:

$P\left(A\right)\times P\left(B\right)$ does not equal 0.4

Instead of using $P\left(\text{A and B}\right)=0$ or $P\left(\text{A and B}\right)=P\left(A\right)\times P\left(B\right)$ as a formula, we can also use it as a test to see if such events satisfy the conditions to be a Mutually Exclusive Event or Independent Event.

{S11-P63} Question 4:

Tim throws a fair die twice and notes the number on each throw. Events A, B, C are defined as follows.

A: the number on the second throw is 5

B: the sum of the numbers is 6

C: the product of the numbers is even

By calculation find which pairs, if any, of the events A, B and C are independent.

Solution:

Probability of Event A = $\times$ *P(Any Number) P(5)*

$$\therefore P\left(A\right)=1\times\frac{1}{6}=\frac{1}{6}$$

Finding the probability of Event B

Number of ways of getting a sum of 6:

5 and 1 1 and 5 4 and 2 2 and 4 3 and 3

$$\therefore P\left(B\right)=\left(\frac{1}{6}\times\frac{1}{6}\right)\times 5=\frac{5}{36}$$

Finding the probability of Event C

One minus method; you get an odd only when odd multiplies by another odd number:

$$1-P\left(C\right)=\frac{3}{6}\times\frac{3}{6}$$

$1-P\left(C\right)$ *represents the probability of getting an odd when doing the product of the two numbers, which can only occur when an odd number is multiplied by another odd number which is* $\frac{3}{6}\times\frac{3}{6}$ *. This method is much easier because if one does a direct method, they will have to count Even* $\times$ *Even and Even* $\times$ *Odd.*

Making $P\left(C\right)$ *the subject gives us:*

$$\therefore P\left(C\right)=\frac{3}{4}$$

For an independent event, $P\left(\text{A and B}\right)=P(A)\times P(B)$

$$P\left(\text{A and B}\right)=\ P\left(1\ and\ 5\right)=\frac{1}{36}$$

$$\neq\ P(A)\ \times\ P(B)$$

$$P\left(\text{A and C}\right)=\ P\left[(2,5)+\ (4,5)+\ (6,5)\right]=\frac{3}{36}$$

$$\neq\ P(A)\ \times\ P(C)$$

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$$P\,(\text{B and C}) = \; P\,[(2,4) + \; (4,2)] = \frac{2}{36}$$

$$\neq \; P(B) \; \times \; P(C)$$

$\therefore$ none are independent.

As said from the previous worked question, we use the equation $\text{P (A and B)} = P(A) \times P(B)$ *not as a formula but as a test to see whether two events are independent or not.*

1.4. Independent Events

- Events that are not connected to each other in any way, or the next event does not rely from the previous event.
- Examples of Independent events:
  - Coin tosses
  - Dice rolling
- Examples of *not* Independent events:
  - Picking a ball from a bag (The next probability will increase due to a decrease in the number of balls in the bag)
- Multiplication Law for IEs:

$$P\,(\text{A and B}) = P(A) \times P(B)$$

In other words, the probability that events A and B occur is just by multiplying them.

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